

Q4 2023 Earnings Call

Company Participants

- Mark Murphy, Executive Vice President and Chief Financial Officer
- Samir Patodia, Investor Relations
- Sanjay Mehrotra, President and Chief Executive Officer

Other Participants

- Mehdi Hosseini, Analyst, Susquehanna Financial Group
- Timothy Arcuri, Analyst, UBS
- Tom O'Malley, Analyst, Barclays
- Toshiya Hari, Analyst, Goldman Sachs

Presentation

Operator

Thank you for standing. Welcome to Micron's Fourth Quarter 2023 Financial Call. At this time, all participants are in a listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions) As a reminder, today's program is being recorded.

And now, I'd like to introduce your host for today's program, Samir Patodia, Investor Relations. Please go ahead, sir.

Samir Patodia {[BIO 21103457](#) <GO>}

Thank you, and welcome to Micron Technology's fiscal fourth-quarter 2023 financial conference call. On the call with me today are Sanjay Mehrotra, our President and CEO, and Mark Murphy, our CFO. Today's call is being webcast from our Investor Relations site at investors.micron.com, including audio and slides. In addition, the press release detailing our quarterly results has been posted on the website, along with the prepared remarks for this call.

Today's discussion of financial results is presented on a non-GAAP financial basis unless otherwise specified. A reconciliation of non-GAAP to GAAP financial measures can be found on our website. We encourage you to visit our website at micron.com throughout the quarter for the most current information on the company, including information on financial conferences that we may be attending. You can follow us on X at MicronTech.

As a reminder, the matters we are discussing today include forward-looking statements regarding market demand and supply, our expected results, and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from statements made today. We refer you to the documents we file with the SEC, including our most recent Form 10-K and 10-Q, for a discussion of risks that may affect our future results.

Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, or achievements. We are under no duty to update any of the forward-looking statements to conform these statements to actual results.

I'll now turn the call over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Samir. Good afternoon, everyone.

In fiscal Q4, Micron delivered revenue and gross margin above the midpoint of our guidance, with EPS above the high end of the range. These results reflect our strong execution, and we are well-positioned to drive significant improvements in our financial performance.

We believe pricing has now bottomed. Ongoing demand growth, customer inventory normalization, and industry-wide supply reductions have set the stage for increased revenue, along with improved pricing and profitability throughout fiscal 2024. We continue to expect record industry TAM in calendar 2025 with more normalized levels of profitability.

Fiscal 2023 was a challenging year for the memory and storage industry as the revenue TAM reached a multiyear low, resulting in a significant impact to financial performance. Despite this difficult backdrop, the Micron team stayed focused on our strategy, executed well and accomplished several important milestones.

We achieved record annual automotive revenue, record NAND QLC bit shipments for the full fiscal year, and reached record levels in calendar Q2 for revenue share in data center and client SSDs. We were the first in our industry to introduce 1-beta DDR5 and LP5X DRAM products and the first to ship HBM3E samples with industry-leading performance and power efficiency.

We were also the first to introduce 232-layer NAND SSD products in data center, client and consumer markets. These accomplishments were underpinned by our leadership technology and continued strong progress in manufacturing execution. We achieved world-class mature yields in record time on our industry-leading 1-beta DRAM and 232-layer NAND technologies.

In addition, Micron took several prudent and timely actions to reduce our CapEx and supply in order to address the market imbalances through the course of fiscal 2023.

Our industry-leading technology roadmap continues to progress well. As we have mentioned before, the vast majority of our bits are on leading-edge nodes 1-alpha and 1-beta in DRAM and 176-layer and 232-layer in NAND.

We continue to make good progress on 1-gamma DRAM development using EUV and are on track for production in calendar 2025. Development of our next-generation NAND node is also well on track.

Now turning to our end markets. Customers continued to reduce their excess inventory for memory and storage in fiscal Q4. Most customer inventories for memory and storage in the PC and smartphone markets are now at normal levels, consistent with our prior forecasts. Inventory levels are normal across most customers in the automotive market as well.

Data center customer inventory is also improving and will likely normalize in early calendar 2024. Consequently, we see demand continuing to strengthen, which has led to an inflection in pricing. Some customers have made strategic purchases in DRAM and NAND to take advantage of unsustainably low pricing as the market begins its recovery.

In data center, traditional server demand remains lackluster while demand for AI servers has been strong. Data center infrastructure operators have shifted budgets from traditional servers to higher-priced AI servers. Total server unit shipments are expected to decline in calendar 2023, the first year-over-year decline since 2016. We expect total server unit growth will resume in calendar 2024 to help fulfill ever-increasing workload demand.

We also expect content growth in both AI and traditional servers. Compared to traditional servers, AI training servers contain significantly higher DRAM and NAND content with greater technology complexity, robust product value and higher profitability. We believe our data center revenue has bottomed, and we expect growth in fiscal Q1 and increasing momentum through fiscal years '24 and '25 in our data center business.

Micron has a strengthening portfolio of solutions optimized for bandwidth, capacity and power. These include HBM3E, DDR5 and associated high-capacity modules, LPDRAM, and data center SSDs. This portfolio of industry-leading products positions us well to capture the opportunities presented by data-centric computing architectures and AI.

The introduction of our HBM3E product offering has been met with strong customer interest and enthusiasm. Our HBM3E provides superior bandwidth, power and capacity for generative AI workloads. We developed this industry-leading design using our 1-beta technology, advanced TSV, and other innovations that enable a differentiated packaging solution.

We have been working closely with our customers throughout the development process and are becoming a closely integrated partner in their AI roadmaps. Micron HBM3E is currently in qualification for NVIDIA compute products, which will drive HBM3E-powered AI solutions. We expect to begin the production ramp of HBM3E in early calendar 2024 and to achieve meaningful revenues in fiscal 2024.

Micron also has a strong position in the industry transition to D5. We expect Micron D5 volume to cross over D4 in early calendar 2024, ahead of the industry. We expanded our high-capacity D5 DRAM module portfolio with a monolithic die-based 128 gigabytes module, and we have started shipping samples to customers to help support their AI application needs. We expect revenue from this product in Q2 of calendar 2024.

Last month, we announced the introduction of 128 gigabytes and 256 gigabytes CXL 2.0 memory expansion modules. By leveraging a unique dual-channel memory architecture, we are able to deliver higher module capacity and increased bandwidth. We have shipped samples to several customers and key partners.

In data center SSDs, Micron's entire portfolio utilizes 176-layer or 232-layer NAND in production, a testament to our product and technology leadership. We are well-positioned to serve the growing demand for fast storage as data-intensive AI applications proliferate. We saw strong demand for our data center NVMe SSDs across our AI-focused, industry-leading 30-terabyte product, as well as for our mainstream products.

Micron ended the second calendar quarter with record high revenue share in data center SSDs, based on independent industry assessments. We expect to build on this momentum in fiscal 2024.

In PCs, we continue to forecast calendar 2023 PC unit volume to decline by a low double-digit percentage year-over-year and then grow by a low to mid-single-digit percentage in calendar 2024. AI-enabled PCs will drive content growth and an improved refresh cycle over the next two years.

In fiscal Q4, we saw strong sequential bit shipment growth at PC OEMs driven by demand for LPDRAM in thin client notebooks. We expect to begin revenue shipments of our industry-leading 1-beta-based client D5 in fiscal Q1 to PC OEMs. According to third-party analysts, in calendar Q2, we reached record revenue share in client SSDs for PC OEMs as customers adopted our industry-

leading solutions. Our 232-layer NVMe client SSD is now qualified at large OEMs and shipping in volume production.

Our SSD QLC bit shipment mix reached a new record for the second consecutive quarter, with growth in both client and consumer markets. We continue to expand our footprint in the high-end consumer SSD space with the launch of three new products that extend our reach into professional content creators and enthusiast PC gamers.

In mobile, we expect calendar 2023 smartphone unit volume to be down by a mid-single-digit percentage year-over-year and then grow by a mid-single-digit percentage in calendar 2024. Elasticity, along with a mix shift toward premium phones with greater capacity, is contributing to memory content growth. About a third of smartphones sold today have at least 8 gigabyte of DRAM and 256 gigabyte of NAND, up more than 7 percentage points versus smartphone units a year ago.

Similar to our view on PCs, AI-enabled mobile phones could drive content growth and a stronger refresh cycle over time. Longer term, we see generative AI applications executing on handsets. These applications will continue to drive new requirements for higher capacity, lower power and increased performance in memory and storage.

Last, I'll cover the auto and industrial end markets, which contribute to more stable revenue and profitability. Fiscal 2023 marked another record revenue year for our automotive business. Micron continues to lead in automotive market share and quality. Long term, we expect memory and storage content per vehicle to increase in both ADAS and in-cabin applications. In addition, fast-growing EVs typically contain higher memory and storage content. Our automotive design win trajectory remains strong.

The industrial market showed signs of recovery in fiscal Q4. Inventory levels for memory and storage are stabilizing at distribution partners and at the majority of our customers. We expect the volume recovery that we observed in the second half of fiscal 2023 to continue into 2024. We see strong growth prospects in this market over time, as industrial customers continue to adopt and implement the IoT, AI and machine learning solutions.

As previously discussed, the CAC or Cybersecurity Administration of China decision earlier this year has impacted our business, particularly in the domestic data center and networking markets in China. We remain committed to serving our customers in China for those areas of their business not impacted by the CAC decision. While there is near-term impact to our demand due to these challenges in China, we remain focused on maintaining Micron's global market share.

Our team's grit and Micron's deep relationships with our customers, underpinned by our technology leadership, increasing product momentum, excellent product quality, and extensive manufacturing and supply chain capabilities position us well toward these goals.

Now, turning to our market outlook, starting with demand. We expect calendar 2023 DRAM bit demand to grow in the mid-single-digit percentage range. In NAND, our expectations for demand growth this calendar year have increased from high-single digits to high teens percentage. These are below the expected long-term bit demand growth CAGRs of mid-teens in DRAM and low-20s percentage range in NAND.

While calendar 2023 DRAM demand has been in line with expectations, NAND growth expectations have increased due to stronger than expected demand in certain parts of the consumer market and a trend of greater elasticity in per unit content.

While macroeconomic factors remain a risk, we expect robust year-over-year bit demand growth in calendar 2024 for both DRAM and NAND, driven by improving end-market demand, normalized customer inventory levels, content growth across products, and ongoing growth in AI. Calendar 2024 bit demand growth is expected to exceed the long-term CAGR for DRAM and to be near the long-term CAGR for NAND.

Turning to supply. Significant supply and CapEx reductions across the industry have helped to stabilize the market and are enabling the recovery that is now underway. We see both DRAM and NAND year-over-year supply growth in calendar 2023 to be negative for the industry. We expect Micron's year-on-year bit supply growth to be meaningfully negative for DRAM. We also expect to produce fewer NAND bits in calendar 2023 than in calendar 2022.

In calendar 2024, we expect industry DRAM and NAND supply growth to be below industry demand growth and meaningfully so for DRAM. We believe calendar 2024 is positioned to be a year of recovery in the memory and storage industry. A sustained period of supply growth less than demand growth will strengthen the pace of recovery.

HBM production will be a headwind to industry bit supply growth. Across the industry, the HBM3E die is roughly twice the size of equivalent capacity D5. The HBM product includes a logic interface die and has a substantially more complex packaging stack that impacts yields. As a result, HBM3 and 3E demand will absorb an outsized portion of industry wafer supply. The ramp of HBM3 and 3E production will reduce overall DRAM bit supply growth industry-wide with particular supply impact on non-HBM products as more capacity is diverted to addressing HBM opportunities.

Micron is experiencing a similar impact of our planned HBM3E ramp on our bit supply capability. Micron's bit supply growth in fiscal 2024 is planned to be well below demand growth for both DRAM and NAND, and we expect to decrease our days of inventory in fiscal 2024. We continue to execute to our strategy of maintaining global bit shipment market share for DRAM and NAND while sustaining tight supply and CapEx management discipline.

Micron's fiscal 2024 CapEx is projected to be up slightly compared to fiscal 2023 levels. WFE CapEx will be down again year-over-year in fiscal 2024. We remain focused on carefully managing overall supply growth. In last quarter's earnings call, we communicated that total wafer start reductions in both DRAM and NAND are approaching 30% versus peak 2022 levels. Amid an intense focus on capital efficiency over the last few quarters, we have redeployed a portion of the underutilized equipment to support production ramp of leading-edge nodes in both DRAM and NAND.

Given the higher process step count of these leading-edge nodes, transitioning this equipment results in a significant and structural reduction to our overall wafer capacity in both DRAM and NAND. Due to this structural reduction in capacity, our DRAM and NAND wafer starts will remain significantly below 2022 levels for the foreseeable future.

Our industry supply projections assume a similar structural reduction in wafer capacity industrywide. Lead times to increase this wafer capacity will be long and will depend on improving demand, pricing and financial performance. We expect underutilization to continue in our legacy nodes well into calendar 2024. We see our demand at leading-edge nodes exceeding our supply in fiscal and calendar 2024, particularly in the second half of the year.

Construction CapEx will be elevated to support our plans to build leading-edge memory fabs in Idaho and New York, for which we filed CHIPS applications in August. As we have highlighted before, the requested level of CHIPS grants for our Idaho and New York projects are essential to

the viability and global competitiveness of each of these projects. Our CapEx plans assume that a certain level of CHIPS grant funds will be made available to us in fiscal year 2024.

Assembly and test CapEx is projected to double year-over-year in fiscal 2024, predominantly driven by investments to support HBM3E production. Our planned fiscal 2024 CapEx investments in HBM capacity have substantially increased versus our prior plan in response to strong customer demand for our industry-leading product.

Over the course of calendar 2024, we see accelerating AI-driven opportunities for memory and storage across multiple market segments from the data center to the edge. We are encouraged by the improving industry demand and supply fundamentals. We believe that the CapEx constraints created by the industry profitability environment, coupled with improved inventories, announced supply reductions and the impact of the HBM ramp on DRAM bit supply growth will create conditions that will increasingly tighten the supply-demand balance, particularly in the second half of our fiscal year.

Our Micron team is executing well and they are taking prudent and proactive actions to navigate through the near-term environment and position the company to emerge stronger from the current downturn. We look forward to a recovery in our business financials taking shape in fiscal 2024.

I will now turn it over to Mark for our financial results and outlook.

Mark Murphy {BIO 16136048 <GO>}

Thanks, Sanjay, and good afternoon, everyone.

In the fourth quarter of fiscal 2023, Micron delivered revenue and gross margin higher than the midpoint of the guidance range and EPS above the high end of the range. We are exiting the fiscal year with the business improving due to multiple factors including higher volumes, an inflection in the pricing environment, strong productivity and ongoing capital discipline.

Total fiscal Q4 revenue was approximately \$4 billion, up 7% sequentially and down 40% year-over-year. Fiscal 2023 total revenue was \$15.5 billion, down 49% year-over-year. Fiscal Q4 DRAM revenue was \$2.8 billion, representing 69% of total revenue. DRAM revenue increased 3% sequentially, with bit shipments increasing in the mid-teens percentage range and prices declining in the high-single-digit percentage range. For the fiscal year, DRAM revenue declined 51% year-over-year to \$11 billion, representing 71% of total revenue.

Fiscal Q4 NAND revenue was \$1.2 billion, representing around 30% of Micron's total revenue. NAND revenue increased 19% sequentially, with bit shipments increasing over 40% driven by timing of shipments including strategic purchases and prices declining in the mid-teens percentage range. For the fiscal year, NAND revenue declined 46% year-over-year to \$4.2 billion, representing 27% of total revenue.

Now turning to revenue by business unit. Compute and Networking Business unit revenue was \$1.2 billion, down 14% sequentially. Data center revenue remained weak as customers continued to adjust inventories and as a result of the CAC decision. In fiscal Q1, we expect sequential growth in data center.

Revenue for the Mobile Business unit was \$1.2 billion, up 48% sequentially due to seasonal effects and timing of shipments. Embedded Business unit revenue was \$860 million, down 6% sequentially. Embedded consumer revenue increased sequentially, helped by seasonality while automotive and industrial revenue declined modestly.

Revenue for the Storage Business unit was \$739 million, up 18% sequentially and driven by increased shipments across most of the product portfolio. SBU bit shipments set records for fiscal Q4 and the fiscal year.

The consolidated gross margin for fiscal Q4 was negative 9%, improving seven percentage points sequentially. Gross margin was impacted by lower pricing and underutilization costs, while the sellthrough of previously written-down inventory provided some uplift. For the fiscal year, consolidated gross margin was negative 8%, down 54 percentage points year-over-year driven by price effects, inventory write-downs and the burden of underutilization. Approximately six percentage points of the reduction is from net inventory write-downs.

Operating expenses in fiscal Q4 were \$842 million, down \$24 million sequentially due to ongoing expense-reduction initiatives and the timing of certain R&D program expenditures. For the fiscal year, operating expenses were \$3.6 billion, down \$209 million year-over-year driven by expense-reduction initiatives. On OpEx for the fourth quarter and year, we ended below the target we communicated starting with our September call a year ago. As market conditions improve, we will remain disciplined in all spending, including operating expenses, focusing R&D on the most critical programs and leveraging a competitive and more productive overhead structure.

We had an operating loss of roughly \$1.2 billion in fiscal Q4, resulting in an operating margin of negative 30%, improved from negative 39% in the prior quarter. Fiscal 2023 operating loss was \$4.8 billion, resulting in an operating margin of negative 31%.

We recorded a tax benefit of \$14 million in fiscal Q4, better than expectations and due primarily to lower than expected foreign taxes related to currency effects. For fiscal 2023, total taxes were \$142 million.

The non-GAAP loss per share in fiscal Q4 was \$1.07, compared to a loss per share of \$1.43 in the prior quarter and earnings per share of \$1.45 in the year-ago quarter. Non-GAAP EPS was a loss per share of \$4.45 for the fiscal year.

Turning to cash flows and capital spending. Our operating cash flows were approximately \$250 million in fiscal Q4. For the fiscal year, we generated \$1.6 billion of cash from operations representing 10% of revenue. Capital expenditures were \$1 billion during the quarter and totaled \$7 billion for the fiscal year. This was in line with recent guidance, and for the year, at the low end of the range of estimates we provided on our December 2022 earnings call. Free cash flow was negative \$758 million in the quarter.

Our fiscal Q4 ending inventory was \$8.4 billion or 170 days. As mentioned last quarter, we are holding approximately \$1 billion of strategic inventory stock associated with build ahead of products for cost optimization and risk mitigation. We see days of inventory improving into the first half of the fiscal year and, adjusting for this strategic stock, expect to have only a few weeks of above target inventories as we enter the second half of fiscal 2024. Inventory levels and profitability will remain principal factors in our decisions around wafer starts and capacity planning.

Continuing with the balance sheet, we maintained historically high levels of liquidity. At year-end, we held \$10.5 billion of cash and investments and had \$13 billion of liquidity when including our untapped credit facility. We ended the year with \$13.3 billion in total debt, a weighted average maturity of 2030 on debt, and low net leverage.

Now turning to our outlook for the fiscal first quarter. Demand is improving as customer inventory levels continue to normalize and secular growth drivers remain intact. We expect record DRAM bit

shipments in fiscal Q1. For NAND, we expect fiscal Q1 bit shipments to decline somewhat from fiscal Q4 levels but remain relatively strong.

In China, the Cyberspace Administration of China decision continues to impact our revenue opportunity, and the associated headwind is reflected in our guidance. Fiscal Q1 gross margin is projected to improve sequentially on a greater mix of DRAM and more sellthrough of written-down inventories. We expect approximately 60% of the remaining benefit from lower cost inventories to clear in fiscal Q1. Our gross margin guidance does not contemplate any additional inventory write-downs due to pricing.

Period costs associated with underutilization will weigh on gross margins in the quarter, as first quarter period costs are projected to be similar to the prior quarter. Beyond fiscal Q1, we project gross margin improvement to continue as prices increase and period costs become less of a factor. We expect the rate of price improvement in the second fiscal half to exceed the first half.

We now forecast gross margins to be positive throughout the second half of fiscal 2024. As mentioned last quarter, we expect fiscal Q1 operating expenses to increase sequentially, driven by an increase in R&D and as temporary reductions to employee compensation come to an end. For the full fiscal year 2024, we expect operating expenses to be up by a low-single-digit percentage versus fiscal 2023.

On taxes, we project a material sequential quarterly increase as we move from a credit in Q4 to a more normal expense. As discussed previously, though overall profitability remains low, a minimum level of taxes will occur based primarily on local jurisdiction profit. As we are forecasting a consolidated pretax loss in fiscal 2024, these local factors will drive tax expense again this year.

We estimate our full-year fiscal 2024 taxes to be under \$200 million. A first-quarter tax estimate of \$80 million reflects our forecasted Q1 results in proportion to full-year projected tax expense. Changes in the distribution of profit within the year may result in changes in the tax expense recognized each quarter.

We project our fiscal 2024 capital expenditures to increase slightly versus fiscal 2023 as we balance the long-term capacity needs of the business with ongoing capital discipline and near-term cash flow objectives. Consistent with our comments the last few quarters, we do see WFE CapEx decreasing from fiscal 2023 to fiscal 2024. When factoring higher construction spend and expected grants in fiscal 2024, we forecast our CapEx to be more evenly distributed over fiscal 2024.

A sequential increase in quarterly CapEx, together with improving but still challenging profitability levels in the near term, means free cash flow will remain significantly negative in the first half of the fiscal year. We forecast improved free cash flow in the back half of the fiscal year.

We project our balance sheet to remain strong and net leverage ratio to peak in the second quarter of fiscal 2024. To support the long-term investment priorities of the business, we have ample liquidity and ready access to multiple sources of credit. We will continue to manage our business to maintain financial flexibility and in a manner consistent with our commitment to our investment-grade rating.

With all these factors in mind, our non-GAAP guidance for fiscal Q1 is as follows. We expect revenue to be \$4.4 billion, plus or minus \$200 million. Gross margin to be in the range of negative 4%, plus or minus 200 basis points. And operating expenses to be approximately \$900 million, plus or minus \$15 million. We expect tax expense of approximately \$80 million.

Based on a share count of approximately 1.1 billion shares, we expect EPS to be a loss of \$1.07, plus or minus \$0.07.

In closing, we achieved many successes in fiscal 2023 despite facing a historic downturn. We sustained our technology, product and manufacturing leadership and achieved mature yields in record time on the industry's most advanced nodes in DRAM and NAND. Micron's leading product announcements position us well to address the growing performance requirements of data-centric computing.

In response to severe market conditions, we acted quickly and decisively to cut supply and capital spend, to reduce operating costs and improve productivity, and to maintain a solid and flexible balance sheet. As the business improves in fiscal 2024, we will leverage our strengths in technology, product and manufacturing while maintaining the productivity and capital discipline that we displayed in fiscal 2023.

I will now turn it back over to Sanjay.

Sanjay Mehrotra {[BIO](#) [1977912](#) <GO>}

Thank you, Mark. The past four quarters tested the resilience and agility of our entire industry. While the recovery from this downturn has begun, Micron will exercise continued supply discipline to drive a return to sustained profitability. I am proud of our team's response to adversity, sustaining our technology leadership, improving time to mature yield, and launching a suite of leading-edge products that represent one of the strongest portfolio expansions in Micron's 45-year history.

As our global investment announcements throughout the year clearly show, Micron remains keenly focused on building our business to meet future demand driven by the proliferation of AI from the data center to the edge. I have full confidence in our team, the position we have built for Micron, and our collective ability to capitalize on the opportunities ahead.

Thank you for joining us today. We will now open for questions.

Questions And Answers

Operator

Certainly. One moment for our first question. (Operator Instructions) And our first question for today comes from the line Tom O'Malley from Barclays. Your question, please.

Q - Tom O'Malley

Hey guys, thanks for taking my question. I just wanted to understand the trajectory here on the NAND business. You guys in the quarter kind of took your demand profile from high-single-digits to high teens and you pointed out consumer in particular. What can you do to give us confidence that, that wasn't a pull in from some of your large consumer customers, and that later this year there might be a little hole there, and just talk about the trajectory of where you see that business going, just given you said that it's going to be down sequentially into November? Thank you.

A - Sanjay Mehrotra {[BIO](#) [1977912](#) <GO>}

So with respect to NAND, yes, I mean compared to what we have said before, we saw strong demand, particularly on the consumer, including some parts of the channel and the consumer

part included likes of smartphones, PCs, etcetera, and again, as I pointed out, the channel as well. And keep in mind, with the pricing that has existed for NAND, elasticity has certainly kicked in. The content is continuing to increase in the devices.

Today, flagship smartphones have minimum of 8 gigabyte DRAM and 128 gigabyte of NAND. So that's the overall trend and same thing in PCs that the elasticity is driving, increasing average capacities. And overall certainly strategic vibe have influenced some of the NAND demand for the year as well.

And keep in mind that next year in 2024, we see that the demand growth will be pretty much close to the long-term CAGR for NAND. And the strat vibes, that the strategic vibe that I mentioned, of course, they help improve the inventory position for NAND as well.

So overall, of course, supply cuts have been made in NAND as well. And as we look ahead, we do see that the demand and supply fundamentals will continue to improve on the NAND side as well.

Q - Tom O'Malley

Very helpful. (multiple speakers)

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

And the inflection -- and the inflection in pricing as well has occurred and, you know, particularly in the second half of our fiscal year we would see that continued improvement in the pricing as well.

Q - Tom O'Malley

Helpful. And then just as a follow-up, obviously you're facing headwinds from the CAC ban. Can you just talk about areas of the market where you guys are targeting to help make up some of those bits. Is it just conversations with customers in the consumer space or are you looking at content increases in the data center, just given the puts and takes of you know where you're making up that a double-digit percentage hole and if you see kind of any change to that. Is it getting worse or is it getting better into February and May of this year? Thank you very much.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

So as we have mentioned before, the CAC headwinds are primarily in the data center and networking markets for us in China. And the impact, the negative impact of revenue as a result of CAC decision is already baked in in our CQ4 results, and is also included in our CQ1 -- FQ1 guidance here.

And keep in minds that the CAC decision, I mean, continues to remain a risk for our business and the impact in our China demand is meaningful. However, Micron has made strong progress with respect to mitigating the effects as well with our global customers who are not impacted by the CAC decision. And we are mitigating that and effect of the mitigation also is reflected in our FQ4 results, as well as in FQ1 guidance.

So the -- FQ4 results and FQ1 guidance reflect the net effect of the loss of revenue in China as well as the success with some of the mitigation, and of course, we are working on mitigating the China revenue loss with increases in demand for us across our multiple end-markets, across all our global end-markets.

And remember that our goal remains to maintain our global share. While there may be some ebbs and flows in the near term, But you know our goal absolutely remains to maintain our global

market share here in terms of this.

Q - Tom O'Malley

Thank you, Sanjay.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi guys, thank you for the question. I had a gross margin question for you, Mark. You're guiding November quarter up five percentage points sequentially. I was hoping you could provide a bridge, if you will, on a sequential basis. You talked a little bit about you know period costs continuing to be a headwind. You gave a little bit of color for the benefit you'll see from selling through written-down inventory. But, you know what are your thoughts on pricing. What are your thoughts on cost-downs? And if you can give you know directional guidance for FED and the puts and takes there, that would be helpful as well. Thank you.

A - Mark Murphy {BIO 16136048 <GO>}

Sure, Toshi. I -- I am -- so on the fourth to first quarter bridge, as you mentioned, we are forecasting roughly a 500 basis-point improvement. We did say over the last few quarters that -- that this would be the profile of our improvement. I will say that it's gotten incrementally better. And that -- that we said before that our profile would be that we would be positive gross margin in the fourth quarter of fiscal '24. We now believe will be positive gross margin through fiscal '24 and so you'll see this positive margin trend continuing.

As it relates to the first-quarter, we do get a small benefit roughly, you know, roughly a point of incremental low-cost inventory pass-through from what passes through in the first-quarter, which will be around \$600 million versus what we had pass-through in the fourth quarter, which is roughly \$550 million.

We also will see a slight benefit in price indeed in the first quarter. As we've mentioned in the last several quarters, we first saw pockets of improvement. And then, we mention that prices bottoming. We are mentioning on this call that it has bottomed and we are seeing in the first-quarter price improvement and we -- we expect that in this transition period to be somewhat muted in the first half and then pick-up momentum and strength -- strengthened considerably in the second half.

So that's the, you know, general walk for -- for the -- for the first quarter. I will, because there's a lot of puts and takes, maybe just, it's been a very difficult several quarters around both utilization writedowns. And fortunately, we're on the other side of that which we can cover later in this call or the after call.

Q - Toshiya Hari {BIO 6770302 <GO>}

Thank you. And then maybe as a quick follow-up on HBM for Sanjay. You know, three months ago, you guys talked about or at least hinted that in fiscal '24, you might be able to generate you know several \$100 million in revenue. Is that sort of target still intact? Have things improved since then and I guess at what point do you expect your presence in HBM to be similar to your presence in DRAM overall? Thank you.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

We are very excited with our HBM product. It is an industry-leading product with respect to performance, power, capacity, capability. And as we have mentioned, this product is in the qualification stages with our customers here and we expect revenue to begin in early 2024. And yes, we are very much still on track for meaningful revenue, \$700 million in our fiscal year '24.

So pleased with the progress, continuing to make good progress. It is an exciting opportunity for the memory industry and Micron will be well-positioned to capture the generative AI opportunities that require the kind of attributes that our HBM3E memory brings to the market.

And, of course, as we proceed through the fiscal year, we expect to be gaining share in this important part of the high -- in this important high growth part of the memory market.

Q - Toshiya Hari {BIO 6770302 <GO>}

Thanks so much.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks a lot, Mark, I wanted to clarify the statement on gross margin. So, you had said in the prepared remarks that it would be positive in the back half of the fiscal year. But in response to the question just before, you said that it will be positive through fiscal '24. So since you're guiding negative in fiscal Q1, does that tell us that fiscal Q2 gross margin will be positive?

A - Mark Murphy {BIO 16136048 <GO>}

Good question, Tim. I didn't -- I didn't comment on fiscal Q2. I -- fiscal Q3, we believe, will be positive. And there'll be a trend of improving gross margin fiscal Q1 through fiscal Q3.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Okay. So you don't want to -- you don't want to state that fiscal Q2 will be positive, maybe it will be, maybe it won't.

A - Mark Murphy {BIO 16136048 <GO>}

It -- maybe. It depends on of course pricing and we're continuing to drive productivity and we'll just have to see at these profitability levels, mix and other things matter. I mean, we do have favorable mix of DRAM going into the first half. You know, we're increasing DRAM relative to NAND in the first half of the year. So that helps.

You know we -- you know, pricing, the momentum is definitively positive. It's just a case of you know we're in a transition period, we did see you know, we kind of had to meet the market on some special deals here in the fourth quarter and you know those -- those effects will continue in the first half, but you know we're definitely seeing new deals being struck at higher, higher prices and we certainly expect that to strengthen in the second-half.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Got it, thanks. And then as a follow-up, just on that comment you just made about called special deals and -- that you, you know, struck in, you know, fiscal Q4. NAND demand, you're more positive on NAND demand, but pricing was still as bad if not worse than at least I thought it would have been in -- you know, side of fiscal Q4. So is -- can you kind of foot that, is that maybe you offering some big deals on the NAND side, some you know customers coming in and being opportunistic and you offerings some big deals to move bits and maybe shore up your NAND share, but at the expense of price. Can you just -- can you just walk through that force? Thanks.

A - Mark Murphy {[BIO 16136048](#) <GO>}

Yeah. I would say, Tim, that, you know, really NAND pricing was not -- was basically in line with the downs that we saw in the third quarter. So, I would say that prices -- the decrease had slowed and the price or the market was being the firm off. You know that -- you know customers realize that this is a period that's ending, the markets firming up, you know, and -- and -- and so, you know, there is also some mix effects in there.

You know, we -- we do see price in NAND improving in the first quarter. It will be up actually in the first quarter. And -- and so I think that sets [ph]odd sound price for NAND.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Thanks, Mark.

Operator

Thank you. One moment for our next question. Our next question comes from the line of Mehdi Hosseini from Susquehanna Financial Group. Your question, please.

Q - Mehdi Hosseini {[BIO 4362002](#) <GO>}

Yes, sir. Thank you for taking my question. Two follow-ups. I would like to better understand the dynamics of seasonality. You talked about the NAND bit shipment declining in the Q1 fiscal year and most of the growth coming from DRAM, but how should we think about seasonality impact in February quarter? And I have a follow-up.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

So certainly, the typical seasonality will be in place. But just keep in mind that FQ4, we had a very strong sequential bit growth in the NAND business. And -- I mean coming off that big sequential growth, we are just guiding to you know normalized level of overall shipments as part of FQ1 and overall, you know -- and again, the content continues to increase across the end-market applications. And that will continue to drive increased demand growth for us as well.

Q - Mehdi Hosseini {[BIO 4362002](#) <GO>}

But would there be seasonality in February quarter?

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

Well, typical seasonality trends that exist in the consumer market would be there. But keep in mind that you know it all depends on the mix of the business as well for us. In terms of NAND, you know, in -- it -- maybe I have already guided to the trends. But overall when we look at the full-year basis, we would look at you know that calendar year '24, NAND will be growing in line with the long-term CAGR or near the long-term CAGR and DRAM will be growing in terms of demand

much ahead of the long-term CAGR as well and all of that will of course take into account the typical seasonality that -- that occurs in the industry.

Q - Mehdi Hosseini {[BIO 4362002](#) <GO>}

Great, thanks for detail. And second question has to do with the puts and takes in reducing wafer starts. And I understand that more emphasis is on the trade mix. But as you -- as I think about memory like, there is no trailing edge. And in that context as you think about bringing utilization rate back up to the normal level. and some of the trailing edge converted to the leading edge, could that help with a bigger step-up in gross margin improvement? How should we think about it, assuming that the trailing edge would be phased out?

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

Well, good question and I think I would like to take the opportunity to provide some context here and overall background. As you know that in 2023, the industry has experienced extreme oversupply and you know extreme negative effect on the profitability as well. And you see now that CapEx cuts and underutilization in the fab have been implemented across the industry given the CapEx constraints that we have, as well as given the poor profitability and certainly Micron has done that, but this is happening across the industry as well.

And at the same time, the demand for the new products is increasing that requires, you know, as you were pointing out, leading-edge technology as well. And in order to maintain our supply discipline and to meet the demand for these new products such as HBM, such as DDR5, we are shifting some of our equipment from older nodes into the newer technologies to ramp up those newer technologies into production.

In the past, we would have done this with more CapEx. But we are being extremely mindful of CapEx spend, extremely disciplined about supply. So as a result, when we move the equipment from older nodes to support the ramp-up of leading-edge nodes, it results in a net reduction in overall -- in a net reduction and the structural reduction in the wafer capacity as well. And that overall bodes well for the industry demand-supply fundamentals.

So keep in mind that as we go through the year, we -- and you know, as we shift more of the equipment toward ramp up of leading-edge nodes, this will result in overall wafer capacity reduction and lowering underutilization as well. Of course, the legacy nodes will continue to have underutilization through the course of the year. And these are all helpful factors in terms of the demand-supply balance in the industry.

And you know, some of the phenomena that we have described regarding structural wafer capacity reduction, this is not unique to Micron. We believe this is happening with other suppliers as well as they grapple with the same supply profitability and CapEx considerations as we are adjusting.

So the reduction in wafer capacity is actually the headwind and supply growth. And this you know that increasing wafer capacity has a high bar for auto -- for CapEx investments, it has a high bar -- bar in terms of ROI, and of course, it would require longer lead-time as well given the new equipment that would be required. So reduction -- structural reduction in wafer capacity is overall good coupled where the underutilization in the legacy nodes.

And as we pointed out, the new products in the leading-edge nodes also do require more wafer capacity for the same gigabits of production given the nature of like HBM that -- where the die-size is twice as big. So, these new products also actually have a favorable impact on the industry

supply growth capability. And these trends of supply growth with respect to structural wafer capacity-reduction and the new products that end up requiring more new wafer capacity are extremely important factors in terms of really understanding the improving, strengthening demand-supply fundamentals as we look at our fiscal year '24 and calendar year 20 -- 2024.

Q - Mehdi Hosseini {[BIO 4362002](#) <GO>}

Thank you.

Operator

Thank you. This does conclude the question and answer session, as well as today's program. Thank you, ladies and gentlemen, for your participation. You may now disconnect. Good day.

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